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Assessment of Factors Affecting Carbon Dioxide Emissions from Power Generation in the United States: a Decomposition Analysis

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Abstract

The additive logarithmic mean Divisia index (LMDI) method of complete decomposition is used to examine the role of four factors (production, energy-mix, intensity and emission factor) affecting the evolution of CO2 emissions from power generation in the United States (US). The US is responsible for little over one-sixth of the world's energy-related CO2 emissions and about one-fifth of world's CO2 emissions from power generation in 2009. The analysis shows production effect as the major factor responsible for rise in CO2 emissions both in the past (1990-2005) and future (2005-2030). On the contrary, both energy-mix and energy intensity effects are key factors responsible for decrease in emissions during these periods. However, the degree of influence of these factors affecting changes in emissions varies between sub-periods. The analysis also shows that there is a potential of efficiency improvement of fossil-fuel fired power plants and its associated co-benefits including energy security and sustainable environmental development. Furthermore, this study finds that the influence of enactment of American Recovery and Reinvestment Act (ARRA) in 2009 on power sector CO2 emissions in the future is minimal.

Keywords: *Power generation; CO2 emissions; Decomposition analysis; ARRA; USA*

Introduction

There is growing scientific evidence and increasing public concern of an increase in greenhouse gas (GHG) concentrations in the atmosphere contributing to rising global temperatures and to changes in climate patterns. The primary source of this increased atmospheric GHG concentrations has been the rising fossil-fuel use and other human activities. Since 1750, it is estimated, about two-thirds of anthropogenic CO2 emissions the most important anthropogenic GHG have come from fossil-fuel burning; and in recent years these emissions have continued to increase. A recent study the United States Environmental Protection Agency (US EPA) finds that increasing GHG emissions may endanger public health or welfare and these emissions are a serious problem for both present and future generations [1]. Based on the scientific study by Solomon et al. [2], stabilizing GHG concentration in the range of 445 490 parts per million (ppm) of CO2-equivalent would require these emissions to be reduced by 13% lower than 2005 emissions.

The US is one of the major sources of energy-related CO2 emissions in the world. In 2009, the US is the second largest source of energy-related CO2 emissions in the world , next only to China, producing about 18% of the world total [3]. These two countries together accounted for about 42% of world's CO2 emissions in 2009. The next three largest sources (India, Russia and Japan) produced only an additional 15% of the world total. Despite its significant share in global CO2 emissions, the US government has signed but not ratified the Kyoto Protocol a legal binding commitments for the reduction of GHGs produced by industrialized nations. The US's non-ratification of this treaty also hinders an international effort in engaging major GHGs emitting developing countries, including China and India, to take on post-Kyoto commitments. In recent years, however, several domestic GHG reduction initiatives, including market-based cap and trade program and sectoral approaches-based commitments, are underway in the country. Although it failed to secure the votes needed to close debate on the bill and move to a final vote, Lieberman-Warner Climate Security Act of 2008 (S.3036) a detailed bipartisan bill that required mandatory and economy-wide GHG emissions reductions through cap and trade program, has become a topic of discussions in the country. Also, the enactment of American Recovery and Reinvestment Act (ARRA) in mid-February 2009 a financial package to stimulate investments in energy efficiency and renewable energy is now one of the centerpieces of federal policy towards mitigating CO2 emissions.

Power generation which includes both electricity and heat generation is one of the major sources of energy-related CO2 emissions in the country. For example, power generation is responsible for 41% (2.2 Gt) of country's total energy-related CO2 emissions in 2009 [4]. Furthermore, the power sector is likely to play an important role in economy-wide cap and trade initiatives in the country toward control of CO2 emissions. Therefore implications of power generation on mitigation of CO2 emissions have become an increasingly important topic for researchers and policymakers. However, establishing any effective policies and measures to reduce CO2 emissions from this sector requires understanding of trends and factors affecting these emissions. Many factors influence these emissions, including socio-economic developments, structural (fuel-mix) and technological changes, and institutional frameworks. Despite the policy relevance of the discussion, very few studies have attempted to address these issues in the context of power sector in the US using quantitative methods. For example, a study by Morgan et al. [5] explored electric power industry's options for reducing its GHG emissions over the next half century covering mainly technological aspects, while others [613] studied these factors in general either focusing on economy-wide energy issues or CO2 emissions in the US. This paper examines as to how these factors are affecting the changes in CO2 emissions from power generation in the US in the past (19902005) and future (20052030) periods by applying a quantitative decomposition method. In addition, this study analyses implications of thermal efficiency improvement on fuel savings and corresponding CO2 emissions reductions.

In recent years, decomposition analysis technique provides widely used analytical tools for studying the factors affecting changes in energy and gas emissions. There are two commonly used decomposition analysis techniques in the literature: Index Decomposition Analysis (IDA) that uses aggregate data at the sectoral level and Structural Decomposition Analysis (SDA) that uses input-output tables. Among the different IDA techniques, Ang [14] recommended Logarithmic Mean Divisia Index (LMDI) method over other IDA methods because this method has several desirable advantages including time independence, ability to handle zero values and consistency in aggregation. Few recent studies [1518] raised the issues of negative values and large number of zero values in the data set in the application of LMDI method. However, an analysis by Ang and Liu [19] shows that replacing zero values with sufficiently small positive value (e.g., -150) would lead to satisfactory decomposition results. In the case of negative values, Ang and Liu [20] proposed a procedure to deal with negative values for the LMDI method. In this paper, the IDA technique using additive LMDI method adapted from Ang [21] and OEE [22] is used.

The paper is divided into six sections. Section 2 discusses a brief overview of power generation and CO2 emissions in the US over the period 1990-2030. Section 3 describes the methodological approach. Section 4 gives a brief description of data and data sources. Analyses of results and key findings are presented in Section 5. The final section provides the conclusion.

Overview of power generation and CO2 emissions in the Unites States

The CO2 emissions is the primary anthropogenic GHG emitted in the US representing over 83% of total GHG emissions (6.6 Gt CO2-equivalent) in 2009 [4]. Combustion of fossil-fuels is the largest source of these CO2 emissions in the country accounting for about 95% of the total. In 2009, the US is also the second largest emitter accounting for 18% of global CO2 emissions from fuel combustion after China (24%). The next three largest emitters are India (5.5%), Russia (5.3%) and Japan (3.8%). These top five emitting countries are responsible for more than half (57%) of global CO2 emissions in 2009 [3].

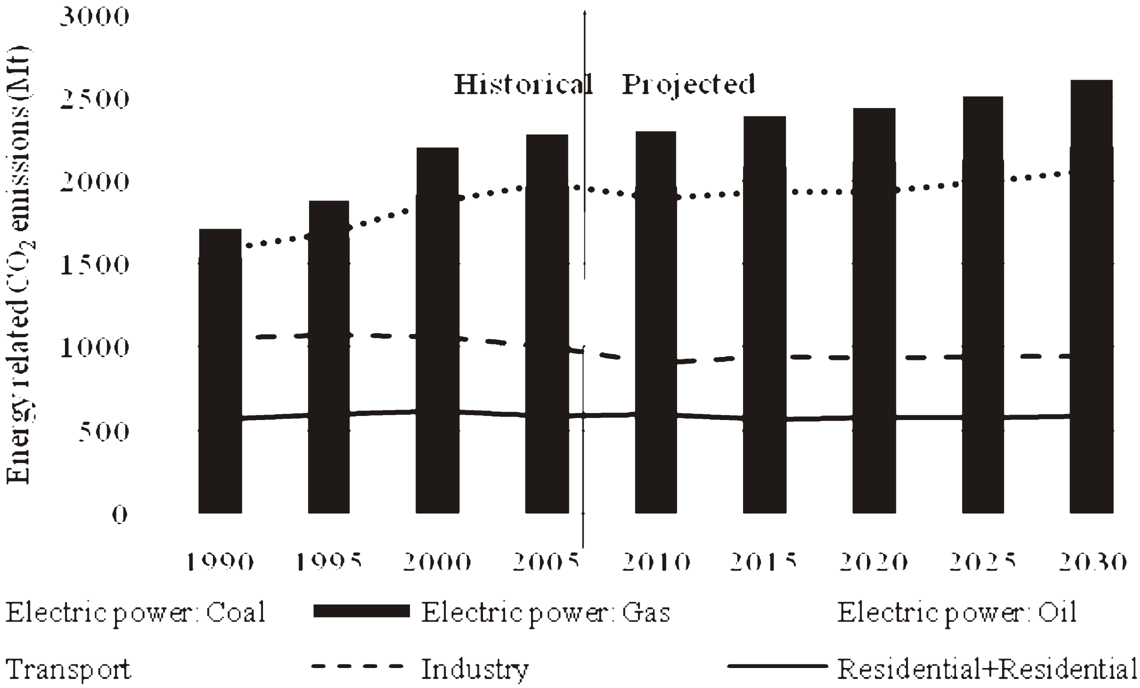

Fig. 1. Sector-wise energy-related CO2 emissions in the United States for the period 1990-2030.
Sources: [23, 24].

Sector-wise, power generation has been the main source of CO2 emissions from fossil-fuel combustion in the country. For example, this sector is responsible for 40% (2.4 Gt CO2) of energy-related CO2 emissions in 2005, followed by transport (33%), industry (17%) and residential and commercial combined (10%) (Fig.1). Between 1990 and 2005, power generation experienced the fastest rate of CO2 emissions growth by an average 1.9% per year compared to 1.5% per year in transportation, 0.3% in residential and commercial sectors combined and negative 0.3% in industrial sector. Furthermore, this sector is projected to remain the main source of energy-related CO2 emissions in the future accounting for about 43% of the total emissions in

2030. Amongst the fossil-fuels, coal remains the dominant fuel responsible for 82% (2.0 Gt CO2) of energy-related CO2 emissions in 2005 and if the current trends continues this share is projected to increase to 83% (2.2 Gt CO2) in 2030 [24]. Over the next 25 years, the share of gas in total energy-related CO2 emissions is estimated to increase from 13% in 2005 to 15% in 2030 while the share of oil is estimated to decrease from 4% in 2005 to 2% in 2030 (Fig.1). The increase in CO2 emissions

Table 1: Overview of historical and projected power generation and corresponding CO_2 emissions in the United States

		Historical		No-stimulus		Stimulus	
		1990	2005	2020	2030	2020	2030
Total power generation	TWh	2904	3927	4374	4809	4358	4744
Of which fossil-fuel-fired power generation	%	69	71	67	68	65	67
Total energy-related CO_2 emissions	Mt	5019	5975	5931	6244	5905	6207
Of which power sector	%	36.3	40.1	41.8	42.4	41.5	42.3
Avg. thermal efficiency of power plants	%	32.9	34.2	34.5	35.4	34.1	35.1
Of which coal-fired power plants	%	33.0	32.8	32.9	33.1	33.0	33.1
Of which oil-fired power plants	%	31.7	32.1	29.9	29.9	29.9	29.8
Of which gas-fired power plants	%	32.7	39.5	40.4	43.1	39.0	42.3
Per capita power generation	kWh/capita	11639	13286	12769	12836	12722	12662
Per capita total CO_2 emissions	t/capita	20.1	20.2	17.3	16.7	17.2	16.6
Power sector CO_2 emissions intensity	kg/US$	0.256	0.218	0.162	0.131	0.159	0.132

Sources: [23—27].

from power sector is closely associated with increase in power generation in the country. Between 1990 and 2005, the total power generation in the US is increased by 35% from 2904 TWh in 1990 to 3927 TWh in 2005 and it is estimated to increase to 4809 TWh in 2030 (Table 1). Of the total power generation, fossil-fuel fired power generation shared more than two thirds of the total generation followed by nuclear (20%) and renewable (8%) energy sources in 2005. Less than 1% of total power generation is imported from Canada in the same year. In terms of average annual growth, the total power generation in the country grew by 2% per year over the past 15 years while it is estimated to grow by only 0.8% per year in next 25 years driven mainly by fossil-fuel fired power plants. Several factors contribute to slower growth of power generation in the future compared to past 15 years, including weak economic growth, improvements in thermal efficiency of fossil-fuel fired power plants and an increase in renewable energy share in total power generation. For example, the US gross domestic product (GDP) is projected to grow only by average 1% per year during the period 20052030 compared to average 2.3% per year in the past (19902005) while the share of renewable energy in total power generation is projected to increase to 12% in 2030 from 8% in 2005. In addition, the improvement in overall average thermal efficiency of fossil-fuel-fired power plants is estimated to be modest at 0.12% per year during the period 2005-2030 (Table 1). More specifically, the efficiency improvement varies from 0.03% per year in coal-fired plants to 0.34% per year in gas-fired plants and to negative 0.29% per year in oil-fired plants during the period 2005-2030. The reason for decreasing efficiency of oil-fired plants is mainly due to the plan of no additional new oil-fired plants and the continuation of existing oil-fired power plants over the period 2005-2030.

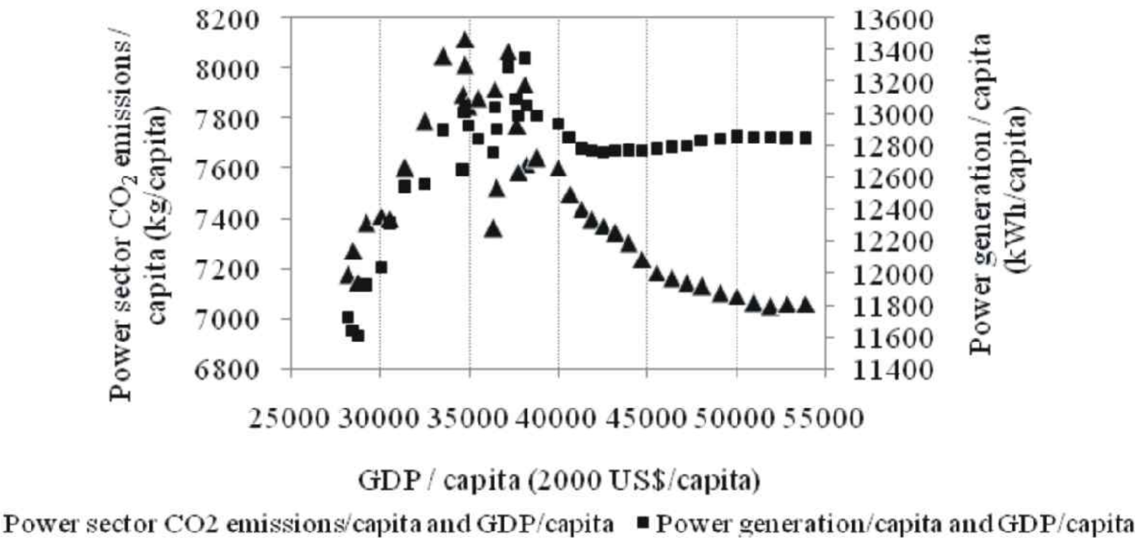

Fig. 2. Relation between income, power generation and CO2 emissions from power in per capita terms in the United States for the period 1990-2030.
Sources: [23–27].

In per capita terms, GDP grew by an average 1.8% per year and power generation grew by an average 0.9% per year over the past 15 years. It is projected that per capita GDP to grow much less at an average 1.5% per year while per capita power generation to grow negative 0.1% per year during the period 2005-2030. This suggests that unlike in the past, the rise in per capita GDP in the country does not lead to increase in per capita power generation during the period 2005-2030 (Fig.2). In contrast, per capita power sector CO2 emissions grew by an average 0.7% per year in the past and it is projected to increase much less at 0.3% per year during the period 2005-2030. The high level of CO2 emissions per capita in proportion to its higher GDP per capita in the past suggests that the power generation in the US is more carbon-intensive compared to decline in per capita power generation CO2 emissions in proportion to higher per capita GDP during the period 2005-2030 (Fig.2).

Methodology

Following the IPCC [28] method, the total CO2 emissions from power generation (C) is estimated by:

$$C = \sum_i C_i = \sum_i FC_i \times EF_i \times O_i \qquad i = 1,2,3,....i,...,n \tag{1}$$

where C_i is the CO_2 emissions from power generation from energy source i, FCi is the amount of energy i used, EFi is the CO_2 emission factor of energy source i and O_i is the fraction of carbon oxidized of energy source i.

The IDA identity for evaluating CO_2 emissions from power generation is given by:

$$C = \sum_i C_i = \sum_i G \frac{G_i}{G} \frac{Q_i}{G_i} \frac{C_i}{Q_i} = \sum_i G s_i f_i u_i \, , \tag{2}$$

where G and G_i are the total power generation and generation from energy source i respectively, Q_i is the energy input to power generation using energy source i, $s_i = G_i/G$ is the power generation share of energy source i, $f_i = Q_i/G_i$ is the energy generation intensity of energy source i and

$u_i = C_i/Q_i$ is the CO_2 emission factor of energy source *i*. The energy source i includes both fossil-fuels and non fossil-fuels. Only fossil-fuels emit CO_2 emissions.

Following the complete additive LMDI decomposition without any residual term proposed by Ang [21] and OEE [22], and manipulating Eq. (2), the aggregate change in CO_2 emissions level from year t to year *t+1* (C*tot)* is expressed as:

$$\Delta C_{tot} = \Delta C_{pdn} + \Delta C_{mix} + \Delta C_{ein} + \Delta C_{emf} \ , \qquad (3)$$

where *Cpdn* is the change in total power production effect, Cmix is the change in power generation mix effect, Cein is the change in energy generation intensity effect and Cemf is the change in emission factor effect. In Eq. (3), the production effect (hereafter PDN) represents the change in CO2 emissions due to total power generation, the power generation mix effect (hereafter MIX) shows the change in CO2 emissions due to a structure of generation share of different types of power plants in total power generation (hereafter EIN) and the emission factor effect captures the change of CO2 emissions due to changes in emission factor. Due to data constraints, it is assumed that the CO2 emission factor for each of the fossil-fuels remains unchanged (i.e., Cemf = 0) during the period 20052030 in Eq. (3). The additive LMDI formulae used in the study for three decomposition factors between year t and t+1are:

$$\Delta C_{pdn} = \sum_i L\left(C_i^{t+1}, C_i^t\right) \ln\left(\frac{G^{t+1}}{G^t}\right),$$

$$\Delta C_{gmx} = \sum_i L\left(C_i^{t+1}, C_i^t\right) \ln\left(\frac{s_i^{t+1}}{s_i^t}\right),$$

$$\Delta C_{ein} = \sum_i L\left(C_i^{t+1}, C_i^t\right) \ln\left(\frac{f_i^{t+1}}{f_i^t}\right) \text{ and}$$

$$\Delta C_{emf} = \sum_i L\left(C_i^{t+1}, C_i^t\right) \ln\left(\frac{u_i^{t+1}}{u_i^t}\right),$$

where $L\left(C_i^{t+1}, C_i^t\right) = (C_i^{t+1} - C_i^t)/(\ln C_i^{t+1} - \ln C_i^t)$ is the logarithmic mean function.

Data

In this study, data from several publications from EIA are used. Historical data (19902005) on energy input to power generation and power generation output by each energy sources are based on EIA [23]. The future (20062030) projected data for these variables are based on EIA [24]. For estimating CO2 emissions from fossil-fuel-fired power generation, data on emission factors is based on EIA [29]. The estimated CO2 emissions for both historical and projected future periods from power generation used in the study are slightly lower compared to values reported in EIA [24, 30]. For simplicity, this study excludes non fossil-fuel-based CO2 emissions sources such as municipal solid waste and geothermal that is reported in these two EIA reports. Together, these two non fossil-fuel sources account for less than 1% of total CO2 emissions from power sector over the study period. Historical and future aggregate real GDP are based on BEA [27] and EIA [24], respectively. Likewise, historical and the projected future population of the country are based on USCB [25, 26] and EIA [24], respectively. These variables are used for calculating per capita indices presented in Table 1, Fig. 2 and elsewhere reported in the study. The values of fraction of carbon oxidized used for estimating net CO2 emissions in Eq. (1) are assumed 1.

Results and Discussion

In this section, results of factors affecting the changes in CO2 emissions from power generation in the US are discussed both on temporal and period-wise basis over the historical (19902005) and the future (20052030) periods. Implications of supply-side thermal efficiency improvements on fuel saved and corresponding CO2 emissions and some policy issues are also discussed. Four decomposition factors: activity-, intensity-, generation mix- and emission factor- effects are discussed for the historical period. Since emission factor effect is assumed zero during the period 20052030, only three factors are discussed for the future period.

Historical (19902005) Decomposition Analysis: Fig. 3 shows decomposition factors of CO2 emissions changes in power generation sector with respect to the Kyoto base year 1990. There are significant differences in terms of both magnitude and direction of factors affecting the changes in emissions. In general, there is an upward trend in PDN effect and this effect is responsible for most of the rise in emissions. Conversely, there is a downward trend in EIN effect reflecting the improvement in average efficiency of fossil-fuel fired power plants during 19902005 period. The MIX effect also shows downward trend except in two years (2001 and 2003). The reason for slight upward trend during these two years is because of declining shares of renewable and nuclear energy sources in total power generation. The EMF effect however shows upward trend but this effect is less significant in overall CO2 emissions change. Overall, the total CO2 emissions have risen over the 19902005 period attributed mainly due to PDN effect. Comparing with the US, it is noteworthy that in OECD Europe as a whole, both MIX and EIN effects are responsible for declining emissions while PDN effect is responsible for increasing emissions leading to overall decrease in total CO2 emissions from power generation during the same period [31].

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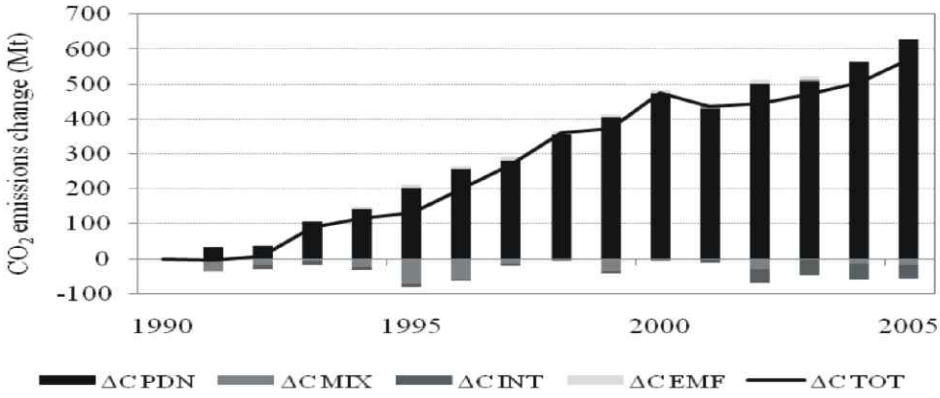


Fig. 3. Decomposition factors of CO2 emissions in power generation sector compared to 1990 emissions level for the period 1990-2005.

Table 2 presents results of both aggregate (19902005) and three 5-year sub-periods (19901995, 19952000 and 20002005) decomposition analysis. These three sub-periods are chosen to reflect the significant changes in CO2 emissions from power generation that took place in the country. The PDN effect is the major factor which would have exceeded the total emissions increase by 110% between 1990 and 2005 in the absence of MIX, INT and EMF effects (Table 2). This increase however is offset by 11% decrease from EIN effect and 9% increase from MIX effect. However, both INT and MIX factors are responsible for decrease in total emissions accounting for 6% and 5% respectively over the same period. The EMF effect contributed about 1% increase of the total emissions.

Table 2: Decomposition factors of CO_2 emissions changes in electric power sector in the United States for the period 1990-2005 (Mt)

	1990-1995	1995-2000	2000-2005	1990-2005
Production effect	202	268	157	629
Energy-mix effect	-72	71	-15	-26
Intensity effect	-8	5	-39	-36
Emission coefficient effect	9	1	-8	5
Total	131	346	95	572

There is a significant change in factors affecting changes in CO2 emissions both in terms of magnitude and direction during three sub-periods. During the first sub-period (19901995), the PDN effect is the major factor accounting for 155% of total emissions increase followed by the EMF effect which accounted for 7% of the total increase. This increase in emissions however is offset by both MIX and INT effects responsible for 56% and 6% decrease respectively during the same period. This decrease in emissions due to MIX effect is because of the combination of increase in gas and carbon-neutral nuclear shares and decrease in coal and oil shares (Fig. 4a). For example, between 1990 and 1995, shares of coal and oil in total power generation decreased from 54% to 52% and 4% to 2% respectively, while shares of gas and nuclear in total power generation increased from 11% to 13% and 20% to 21% respectively. Likewise, the decrease in emissions due to INT effect is because of increase in thermal efficiency gas-fired power plants (Fig. 4b). In contrast, all four effects are responsible for total emissions increase during the second sub-period (19952000). The PDN effect is responsible for most of the emissions increase (78%) followed by MIX effect (21%) and INT effect (1%). In this second sub-period, increased shares of all fossil fuels (coal, oil and gas) and decrease in thermal efficiency of coal-fired power plants are responsible for part of the total emissions increase which is the highest compared to the other two sub-periods (Fig. 4a,b). During the third sub-period (20002005), however, the increase in total emissions is relatively small compared to the first two sub-periods with PDN effect responsible for increase in emissions and the MIX, INT and EMF effects responsible for decrease in emissions. During this sub-period, the net change of decrease in coal share from 53% to 51% and increase in gas share from 14% to 18% in total power generation lead to decrease in total emissions from MIX effect. Between 2000 and 2005, the average thermal efficiency of coal-fired power plants decreased by about 1% while the average efficiencies of both gas- and oil-fired power plants increased from 35% to 40% and 31% to 32% respectively. This net improvement in efficiency is responsible for overall decrease in total emissions from INT effect.

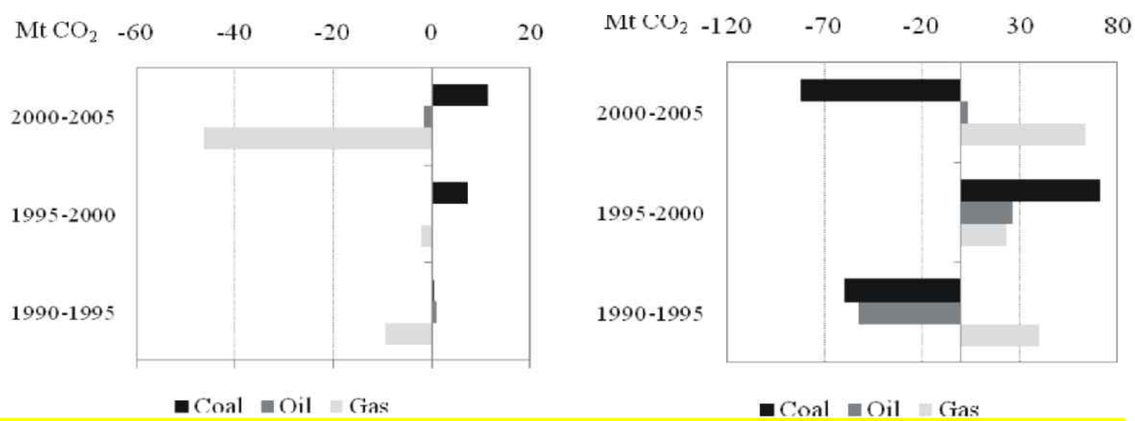
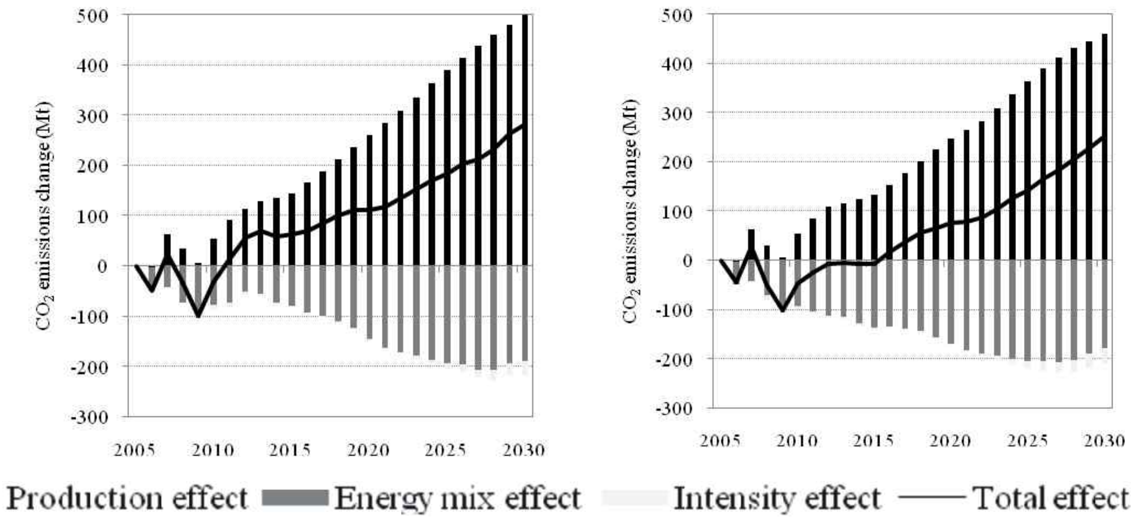


Fig. 4. Changes in CO2 emissions from power generation mix and power generation intensity effects by fuel types for three sub-periods: 1990—1995, 1995—2000 and 2000—2005.

Future Period (20052030) Decomposition Analysis: Table 3 presents the factors affecting the changes in CO2 emissions from power generation in the US for aggregate projected period (20052030) and two projected sub-periods (20052020 and 20202030) under two reference cases. These two reference cases, taken from EIA [24], are the "no-stimulus" case and the "stimulus" case. Broadly, the no-stimulus case is a baseline vision of how the US energy markets are likely to evolve if there are no new energy-policy interventions by the government. However, it takes account of those government laws and regulations that had already been adopted by November 2008. In addition, this no-stimulus case also accounts for several other changes as of mid-February 2009, including updates to macro-economic assumptions and near-term fuel-price projections, reinstatement of the Clean Air Interstate Rule (CAIR) and final action on Corporate Average Fuel Economy (CAFE) standards for model year 2011. In mid-February 2009, the US government enacted the ARRA. The purpose of ARRA is to provide significant new federal funding, loan guarantees and tax credits to stimulate investments in energy efficiency and renewable energy in the country. The stimulus case reflects no-stimulus case combined with the provision of ARRA. Therefore, the effect of ARRA alone, including its energy-related provisions and its stimulus impact on the economy, can be roughly estimated by comparing no-stimulus case to stimulus case.

Table 3: Decomposition factors of CO2 emissions changes in electric power sector in the United States for the period 2005-2030 (Mt)

	No-stimulus case			Stimulus case		
	2005-2020	2020-2030	2005-2030	2005-2020	2020-2030	2005-2030
Production effect	261	243	507	251	215	471
Energy-mix effect	-148	-44	-195	-172	-11	-187
Intensity effect	-7	-33	-40	-3	-29	-33
Total effect	106	166	272	76	175	251

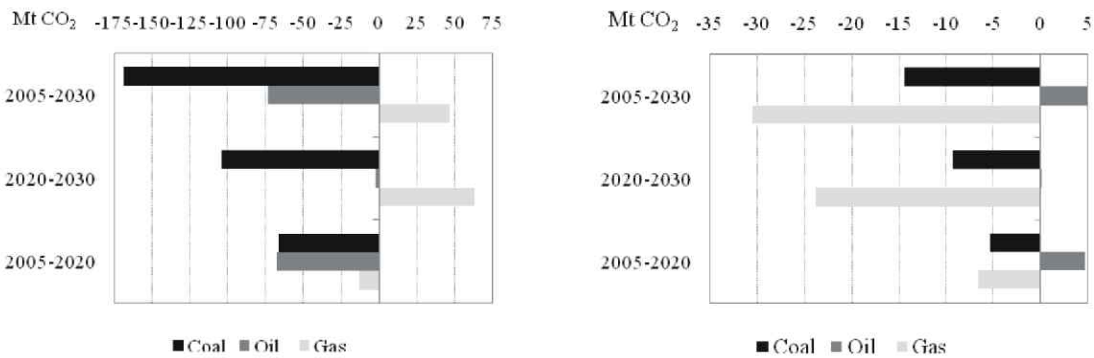


a. No-stimulus case b. Stimulus case

Fig. 5. Decomposition factors of CO2 emissions in power sector compared to 2005 emissions level for the period 2005–2030.

Fig. 5 shows decomposition factors of CO2 emissions changes in power generation sector with respect to year 2005 under no-stimulus and stimulus cases. In both cases, there is an upward trend in PDN effect and this effect is responsible for most of the rise in emissions and there is a downward trend in MIX and INT effects. However, the magnitude of decrease in CO2 emissions from MIX and INT effects combined is smaller relative to increase in emissions from PDN effect leading to overall increase in emissions during the period 20052030. The reason for this downward trend in MIX effect under these two cases is because of declining shares of fossil-fuels (coal, oil and gas) and increasing shares of renewables in total power sector energy consumption. For example, the share of fossil fuels in total power sector energy consumption is estimated to decrease from 71% in 2005 to 68% (no-stimulus case) and to 65% (stimulus case) in 2030 while the share of renewables is estimated to increase from 8% in 2005 to 14% (no-stimulus case) and to 15% (no-stimulus case) in 2030. It is noteworthy that the share of nuclear in total power sector energy consumption is estimated to remain about the same (20%) between 2005 and 2030 under both cases. The INT effect follows downward trend similar to MIX effect but its magnitude is relatively small when compared to MIX effect.

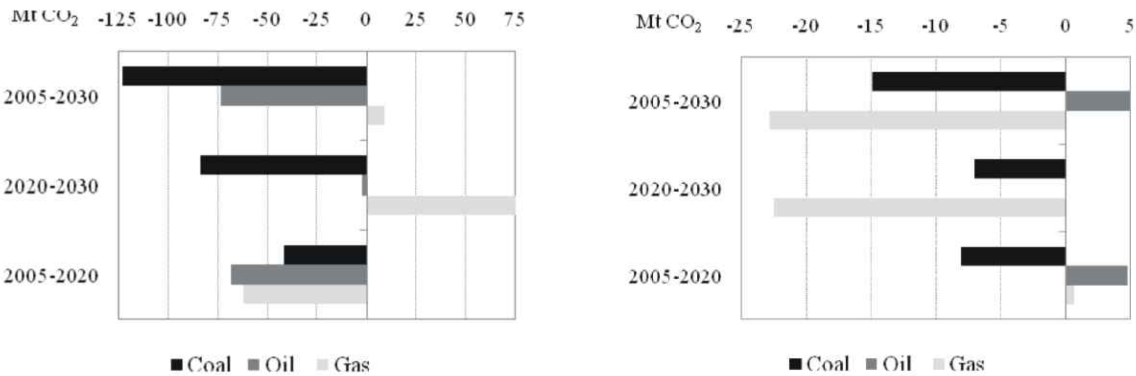
Between 2005 and 2030, the total CO2 emission from power generation is estimated to increase by about 11% under no-stimulus case and 10% under stimulus case. Among the factors, the PDN effect would contribute more than the total emissions increase in the absence of both MIX- and EIN-effects. For example, the PDN effect would be responsible for 186% of the total emissions increase while MIX and INT effects would be responsible for 72% and 15% of the total emissions decrease, respectively, under no-stimulus case during the period 20052030 (Table 3). Likewise, the PDN effect would be responsible for 188% of the total emissions increase and part of this increase is offset by MIX effect (74%) and INT effect (13%) under stimulus case during the same period. Under no-stimulus case, the net decrease in total emissions from MIX effect is attributed to decrease in emissions from coal and oil use (Fig. 6a) while the net decrease in total emissions from INT effect is attributed to improvement in thermal efficiency of coal- and gas- fired power plants (Fig. 6b) during the period 20052030. Under stimulus case, both MIX and INT effects in total emissions decrease follow similar trends in terms of direction and magnitude during the same period (Fig. 7).



a. Energy-mix effect

b. Intensity effect

Fig. 6. Changes in CO2 emissions from power generation mix and power generation intensity effects by fuel types under no-stimulus case for the period 2005—2030.



a. Energy-mix effect

b. Intensity effect

Fig. 7. Changes in CO2 emissions from power generation mix and power generation intensity effects by fuel types under stimulus case for the period 2005—2030.

During the first projected sub-period (20052020), the PDN effect is the major factor accounting for 246% and 330% of total emissions increase under no-stimulus case and stimulus case respectively (Table 3). This increase in emissions however is offset by both MIX and INT effects responsible for 139% and 7% decrease respectively under no-stimulus case and for 226% and 3% decrease respectively under stimulus case. This decrease in emissions due to MIX effect under both cases is because of the decrease in fossil fuels shares in total power sector energy consumption (Fig. 6a and Fig. 7a). Likewise, the decrease in emissions due to INT effect is because of increase in thermal efficiency coal- and gas-fired power plants (Fig. 6b) under no-stimulus case and increase in efficiency of coal-fired power plants (Fig. 7b) under stimulus case. During the second projected sub-period (20202030), these decomposition factors follow similar directions but differ in magnitudes. For example, during the second sub-period, the share of energy-mix effect in total effect is relatively lower while the share of intensity effect is relatively higher compared to first sub-period.

Overall, the decomposition analysis finds that production effect which is closely associated with the growth of economy is the dominant factor for net increase in total emissions for both historical and projected future periods. Conversely, both energy-mix and intensity effects contribute decrease in total emissions over the study period. Decreasing share of fossil fuels together with increasing share of renewables is the main reason for decrease in total emissions due to energy-mix effect while average thermal efficiency improvement of fossil-fuel-fired power plants is the main reason for decrease in total emissions due to intensity effect.

Implications of Thermal Efficiency Improvement and Policy Insights: The imperative to promote energy efficiency remains a priority for the US government. In this regard, thermal efficiency improvement of power plants is seen as one of the attractive options in mitigating energy-related GHG emissions in the country. Specifically, coal-fired power generation is of particular interest from efficiency improvement point of view. This is because coal-fired power plants alone would be responsible for roughly half of the total power generation in the country for next 25 years under both no-stimulus and stimulus cases. Almost all the coal requirement for power generation is domestically available. Furthermore, both higher imported oil and gas prices in recent years and associated energy security concerns are making coal more competitive and attractive as a fuel for power generation in the country. In addition, advance clean coal technologies incorporating CO2 capture and storage (CCS) offer promising CO2 mitigation measure from coal use in power generation in the future.

Table 4: Thermal efficiency improvement of fossil-fuel-fired power plants and its co-benefits in the United States in 2020 and 2030

	Avoided CO_2 emissions (Mt)				Fuel saved (US$ billion, 2007)			
	No-stimulus case		Stimulus case		No-stimulus case		Stimulus case	
	2020	2030	2020	2030	2020	2030	2020	2030
Coal	462	475	466	479	9.5	10.2	9.6	10.3
Oil	16	16	16	16	3.8	4.4	3.8	4.3
Gas	35	19	30	17	4.7	3.0	4.1	2.7
Total	513	510	512	512	18.1	17.6	17.4	17.3

Table 4 presents the potential of CO2 emissions reduction and amount saved from reduced fossil-fuel requirements if the US produce power at the efficiencies observed in Japan in 2005. Wide variations are seen in efficiencies across countries, with OECD countries typically having the highest efficiencies. For comparison purpose, Japan is selected because this country is amongst the most efficient fossil-fuel based power generation in the world. Although South Korea shows higher efficiency of gas-fired plants compared to Japan, we choose Japan for consistency and simplicity. For example, the average thermal efficiency of coal-, oil- and gas-fired Japanese power plants in 2005 are 42.2%, 48.3 and 45.1% respectively [32] while the global average efficiencies of electricity production are 34.3% for coal, 39.5% for natural gas and 36.5% for oil during 2001-2005 [33]. This suggests that an average coal-fired power plant efficiency improvement of almost 9% could be achieved in the US in 2030 if the current best practice technology in Japan is used. This efficiency improvement translates into reduction of total coal requirements by 126 Mtoe under no-stimulus case and by 127 Mtoe under stimulus case. The corresponding CO2 emissions avoided would be 475 Mt under no-stimulus case and 479 Mt under stimulus case (Table 4). More important is the total amount saved from reduced coal demand which would be roughly US$10 billion in 2030. In the case of oil-fired power plants, average efficiency improvement of 18% could be achieved if using the best practice technology in Japan. This corresponds to about 16 Mt of CO2 emissions avoided and about 4.4 US$ billion savings in 2030 under no-stimulus case. Likewise, average natural gas power generation efficiency improvement of 2% could be achieved and this efficiency improvement corresponds to 19 Mt of CO2 emissions avoided and about US$ 3 billion savings in 2030. The potential for energy savings and corresponding emissions reduction would be even higher if using commercially available best practice plants with efficiencies as high as average 47% for conventional coal-fired power plants (ultra-supercritical units) and 60% for natural gas-fired combined cycle plants [34].

The Lieberman-Warner bill mandated a reduction of economy-wide energy-related CO2 emissions of 19% below 2005 level by 2020 which is equivalent of roughly 1019 Mt under no stimulus case. It is noteworthy that efficiency improvement of power plants alone would meet about half of these economy-wide emissions reduction (Table 4). The Group of 8 (G8) leaders agreed at the Heiligendamm Summit in 2007 to seriously consider a global 50% CO2 reduction target. Although it represents a tough challenge, if the US considers reducing it domestic energy-related CO2 emissions by say 20% to 40% from 2005 level by 2030, it would mean roughly 1463 Mt to 2658 Mt of emissions reduction under no stimulus case. Almost one-third (20% reduction) to one-fourth (40% reduction) of this reduction could be achieved through efficiency improvement of power plants.

Table 5: Power generation capacity and net cumulative capacity addition in the United States in 2020 and 2030 (GW)

	Power generation capacity				Net cumulative capacity addition			
	No-stimulus case		Stimulus case		No-stimulus case		Stimulus case	
	2020	2030	2020	2030	2020	2030	2020	2030
Coal	326	336	340	338	14	25	16	27
Oil and Natural Gas Steam	101	99	93	93	-18	-20	-26	-26
Combined Cycle	199	247	180	235	17	66	16	54
Combustion Turbine/Diesel	151	207	141	189	14	74	8	56
Total fossil-fuel-fired plants	776	889	754	855	27	145	14	111

Source: [24].

CCS represents another potential way of mitigating CO2 emissions from fossil-fuel-fired power plants in the country. Although not currently economically feasible, these CCS technologies are expected to play important role in reducing CO2 emissions in the future. This is because many new power plants would be built in the near future to meet increasing energy demand in the country and to replace old plants in the country. For example, fossil-fuel-fired power generation capacity in the US is projected to increase from 734 GW in 2005 to 889 GW under no-stimulus case and to 855 GW under stimulus case in 2030 (Table 5). Between 2005 and 2030, additional 174 GW and 149 GW of new fossil-fuel-fired plants are planned for addition under no-stimulus and stimulus case respectively. Policy measures including legal requirement or financial incentives to encourage and setting market prices of CO2 emission credits could be introduced to motivate developers to make their plants capture ready in the future.

In addition to efficiency improvements and CCS technologies, CO2 emissions reduction from power sector would also require promotion of both low- and zero-carbon technologies in the future. For example, development and deployment of commercially available low-carbon technologies such as coal- and biomass-based integrated gasification and combined cycle (IGCC) generation, coal ultra-supercritical steam and natural gas combined cycles power generation as well as zero-carbon technologies such as nuclear, hydro, wind, solar and geothermal should be promoted. However, barriers related to implementing these technologies such as financial viability, reliability, siting, public acceptability, lack of institutional and regulatory frameworks and other factors also needs to be considered. Furthermore, demand-side end-use energy efficiency improvements, reduction of T&D losses and demand side management (DSM) programs should also be promoted. Analyses of these issues are however beyond the scope of this paper.

Concluding Remarks

The US shares significant portion of global economy, energy use and CO2 emissions. In 2006, this country is responsible for one-fifth of the world's energy-related CO2 emissions and a little less than one-third of world's CO2 emissions from power generation. Between 1990 and 2005, CO2 emissions from power generation increased by almost 32% in the country in contrary to decrease in OECD Europe as a whole.

The decomposition factors of CO2 emissions changes from power generation in the country presented in this study identifies production effect as the major factor responsible for increase in emissions during historical and future periods. On contrary, both energy-mix and energy intensity effects are key factors responsible for decrease in emissions during these periods. But, unlike in the past, energy-mix effect contributes significant decrease in emissions in the future attributed to increasing share of renewables in total power generation. It is also noteworthy that both magnitude and direction of these factors are similar under both no-stimulus and stimulus cases. This suggests that the influence of ARRA for changes in power sector CO2 emissions is minimal. However, the degree of influence of these factors affecting changes in emissions varies between the sub-periods. The historical three sub-periods decomposition analysis shows that the relative magnitude of total emissions increase is the largest during the second sub-period (19952000) than in the first (19901995) and the third (20002005) sub-periods. In this second sub-period, all three factors and the emission factor effect contribute the rise in total emissions. In the future two sub-periods, the second sub-period (20202030) shows larger total emissions increase than in the first sub-period (20052020).

Between 1990 and 2005, shares of coal- and oil-use have decreased while the share of gas use has increased in total fossil-fuel fired power generation. However, coal use in the country remains the dominant fuel in total fossil-fuel fired power generation and it is by far the biggest contributor to incremental emissions during this period. Furthermore, the coal use for power generation continues to remain the dominant fuel and contribute the most in total emissions in the future both under no-stimulus and stimulus cases.

The average thermal efficiency of fossil-fuel-fired power generation in the U.S. is relatively less efficient compared to other industrialized nations. The results find that thermal efficiency improvement of power plants over the next 25 years is one of the attractive supply-side options in mitigating energy-related CO2 emissions and in addressing energy security concern. Associated co-benefits of efficiency improvement include reduced energy demand and CO2 emissions, and financial benefits. CCS represents another potential way of mitigating CO2 emissions from fossil-fuel-fired power plants in the country. Specifically, coal-fired power plant is of particular interest both from efficiency improvement and from application of advance clean coal technologies incorporating CCS use point of view in the country. The other promising measures in containing the growth of CO2 emissions from power generation may include switching to fuels with lower carbon content and the promotion of both low- and zero-carbon technologies including biomass-based IGCC generation, nuclear, solar and wind power. However, new institutional and regulatory frameworks, setting market price of CO2 emission credits, financial schemes and public acceptability are needed to encourage implementation of these climate-friendly technologies.

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